



Building RAG Agents with LLMs

QuCo-RAG: Quantifying Uncertainty from the Pre-training Corpus
for Dynamic Retrieval-Augmented Generation

Jae Y. CHOI

NVIDIA DLI Certified Instructor & University Ambassador

본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

Copyright © Jae Y. CHOI (jaeyeong2022@gmail.com).

This material may be freely used, provided that the author and source are properly credited.

Learning Objectives

Dynamic Retrieval with QuCo-RAG

- Explain when dynamic RAG retrieves
- Follow the two corpus-based checks
- Read the evidence across models
- Recognize the limits of co-occurrence



QuCo-RAG

Quantifying Uncertainty from the Pre-training Corpus

QuCo-RAG: Quantifying Uncertainty from the Pre-training Corpus for Dynamic Retrieval-Augmented Generation

Dehai Min¹, Kailin Zhang², Tongtong Wu³, Lu Cheng¹

¹University of Illinois at Chicago, ²New York University, ³Monash University

dmin10@uic.edu, kz2739@nyu.edu, tongtong.wu@monash.edu, lucheng@uic.edu

From model confidence to corpus evidence

When to Retrieve?

The trigger is a separate decision from the retriever

Static policy

Fixed schedule

Once, or every sentence

Internal signals

Model confidence

Probability · entropy

Corpus statistics

Evidence frequency

Rare entities · unseen pairs

QuCo-RAG changes the trigger—not the retrieval engine

Confident but Wrong

A low-uncertainty token can still be a hallucination



Question: Are the directors of *Il Seduttore* and *The Trial of Joan of Arc* from the same country?

(a) DRAGIN: Token-level Uncertainty via Attention & Entropy

Outputs: *Il* *Seduttore* was directed by Mario Camerini and ...

$Uncertainty = 1.47 > threshold$

High Uncertainty ✗

(Token from the question)

$Uncertainty < threshold$

Low Uncertainty ✗

(Correct: Franco Rossi)

(b) QuCo-RAG: Quantifying Uncertainty via Corpus Statistics

Outputs: *Il Seduttore* was directed by Mario Camerini and ...

Entity Co-occurrence frequency: 0

Potential Hallucination ✓

⌚ milliseconds

Query



Pre-training
Corpus Engine

Wrong target

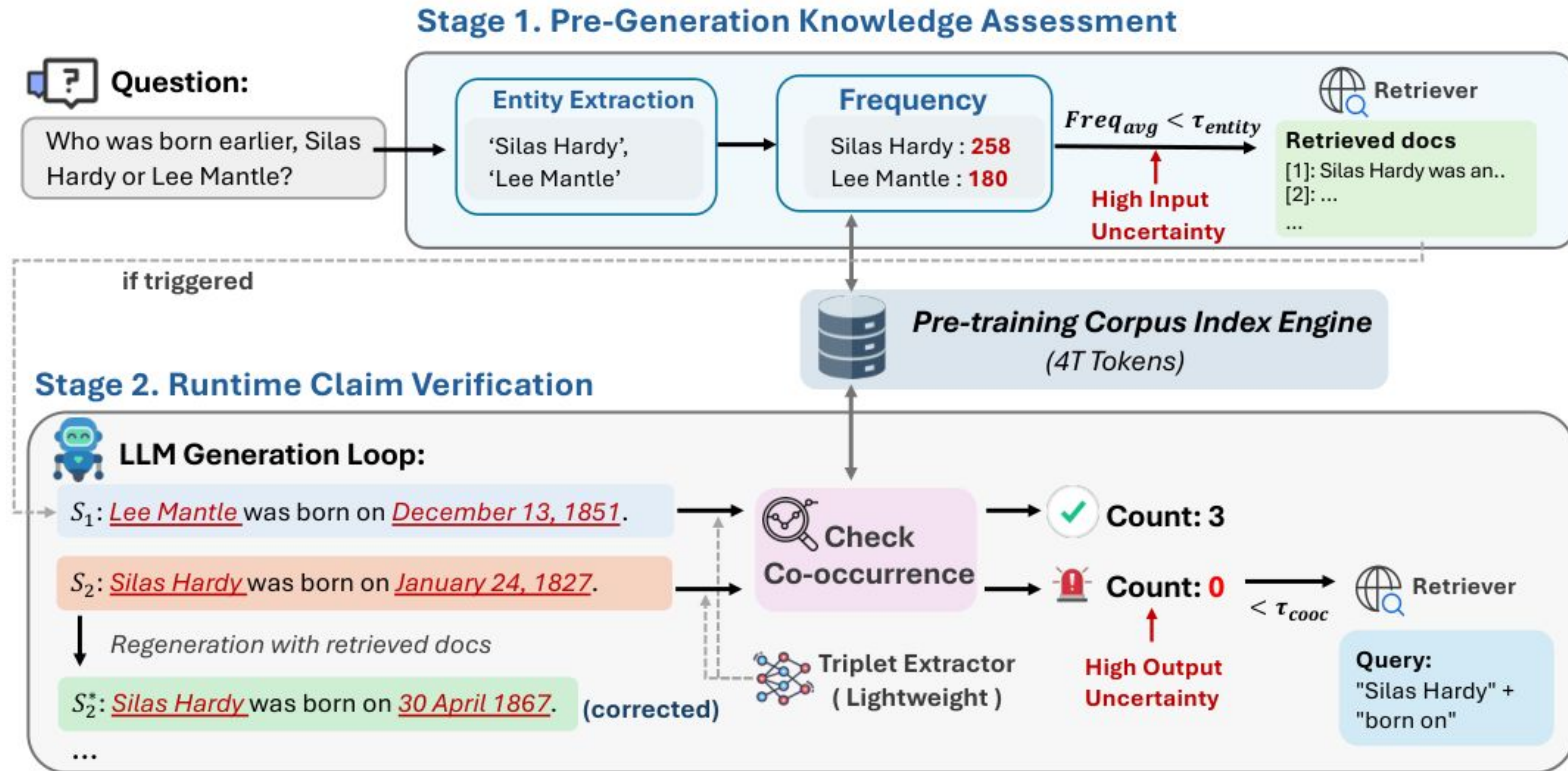
High uncertainty
on “Il”

Missed error

Low uncertainty
on the wrong director

QuCo-RAG Framework

Check before generation. Verify during generation.



Two Corpora, Two Roles

The uncertainty signal and the retrieved evidence are different

Pre-training corpus

$\mathcal{P}$

Infini-gram

Entity / pair counts

Should we retrieve?

Retrieval corpus

$\mathcal{C}$

BM25 + top-3 documents

External evidence

What should the model read?

Stage 1: Before Generation

Is the question about long-tail entities?



$$\delta_{\text{pre}} = \mathbb{I} \left[\text{Avg}_{e \in \mathcal{E}_Q} \text{freq}(e; \mathcal{P}) < \tau_{\text{entity}} \right]$$

Average over
question entities
Threshold: 1,000

Silas Hardy: 258

Lee Mantle: 180

$$\frac{258 + 180}{2} = 219 < 1000$$

Retrieve with the original question

Stage 2: During Generation

Does any entity pair have zero co-occurrence?

Generated sentence $\rightarrow$ (head, relation, tail)

$$\text{cooc}(h, t; \mathcal{P}) = |\{\omega \in \mathcal{P} : h \in \omega \wedge t \in \omega\}|$$

$$\delta_i = \mathbb{I} \left[\min_{(h,r,t) \in \mathcal{T}_i} \text{cooc}(h, t; \mathcal{P}) < 1 \right]$$

Any zero-count pair
 $\rightarrow$ trigger retrieval

h / t: head / tail entities

ω : a 1,000-token window

The relation r is not verified
by entity co-occurrence.

Retrieve and Regenerate

Search for the relation—not the suspected answer

Suspect claim

Silas Hardy was born
on January 24, 1827.

Co-occurrence: 0

Retrieve supporting documents

Semantic query

$$q_i = h \oplus r$$

“Silas Hardy” + “born on”

Regenerated sentence

Silas Hardy was born
on 30 April 1867.

Making the Checks Practical

A small extractor and a searchable pre-training index

Corpus engine

~4T tokens

Infini-gram index

Triplet extractor

0.5B model

40K distilled examples

External retriever

Top-3 passages

BM25 over Wikipedia

No logits or hidden states required from the target LLM



Building RAG Agents with LLMs

QuCo-RAG: How Well Does It Work?

Experimental Setup

Matched-corpus evaluation and cross-model transfer

Questions

2Wiki + HotpotQA

1,000 examples each

EM + token F1

Matched corpus

OLMo-2-Instruct

7B · 13B · 32B

Own training corpus

Proxy corpus

Llama · Qwen · GPT

Same OLMo-2 index

Unknown target corpus

Main Results: OLMo-2

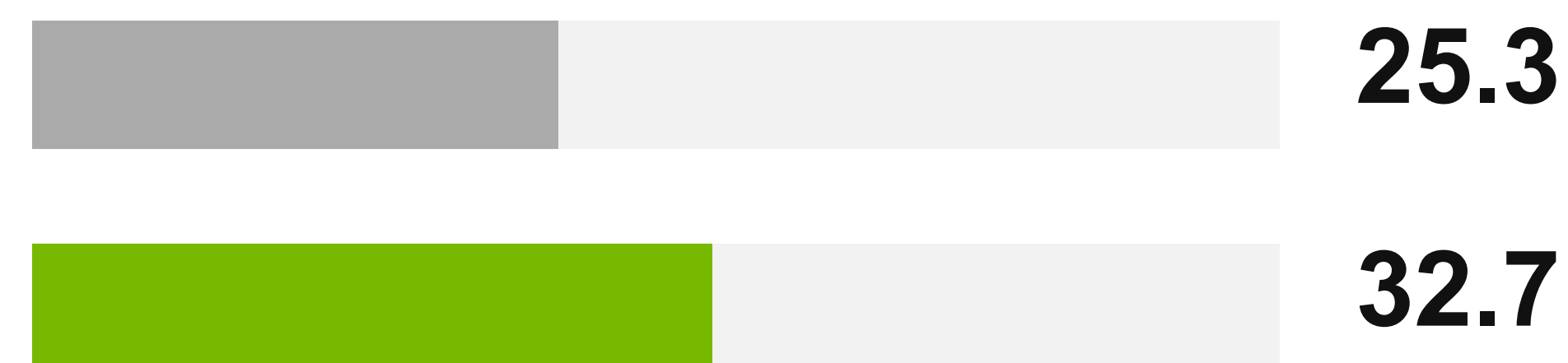
Exact Match (%) · best baseline vs. QuCo-RAG in each setting

■ Best baseline

■ QuCo-RAG

OLMo-2-7B

2Wiki



HotpotQA



OLMo-2-13B

2Wiki



HotpotQA



OLMo-2-32B

2Wiki



HotpotQA



All bars: 0–60% EM · Gains: +5.3 to +12.0 points

Full results → Appendix A1

Transfer to Other Models

The OLMo-2 corpus serves as a proxy—not the exact training set

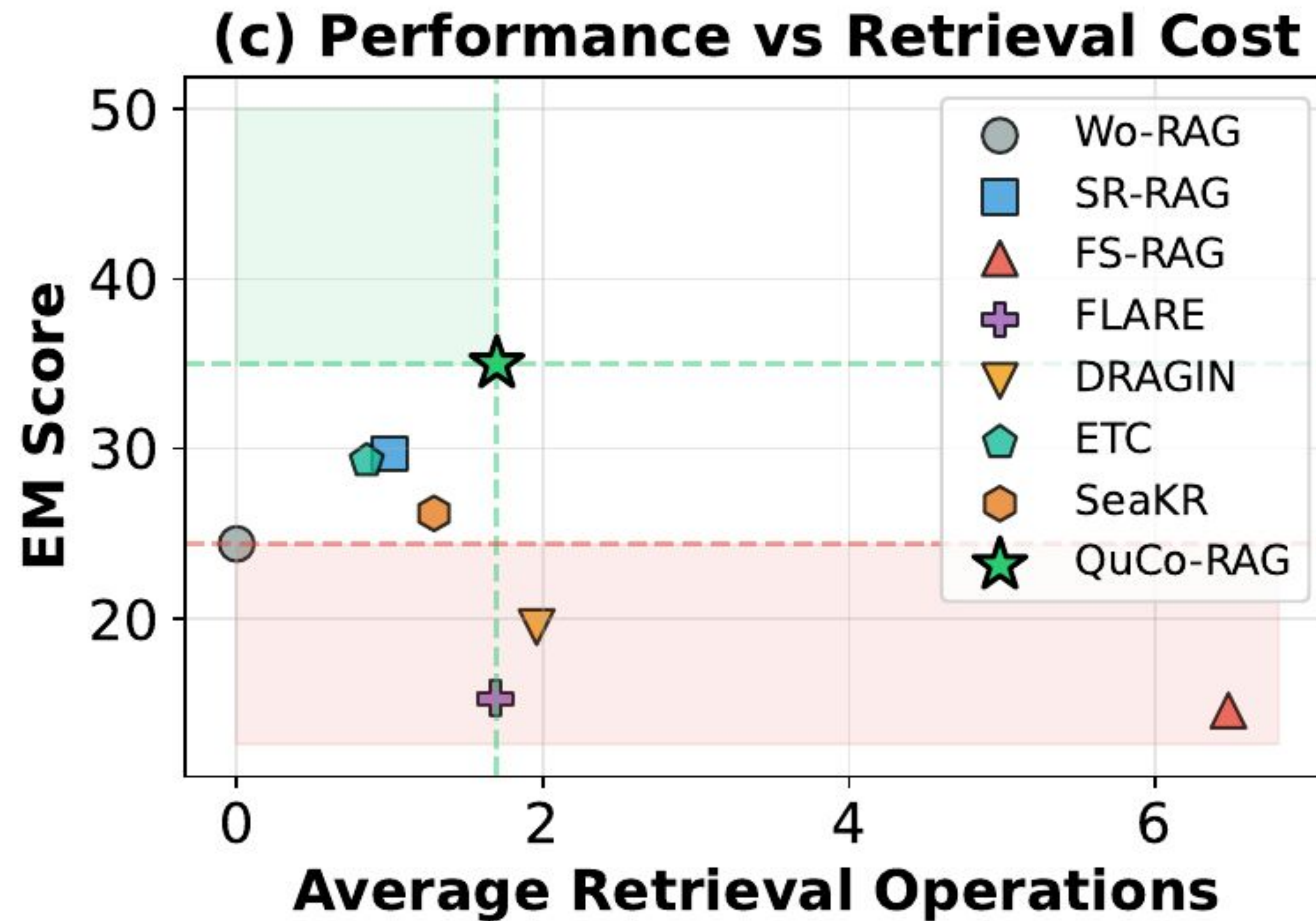
Target model	2Wiki	HotpotQA
Qwen2.5-32B	50.0 (+14.1)	41.6 (+3.0)
Llama-3-8B	38.4 (+4.9)	36.2 (+2.7)
GPT-4.1	64.6 (+4.6)	48.2 (+8.1)
GPT-5-chat	59.7 (+8.7)	48.4 (+5.5)

EM (%) · Parentheses: absolute gain over the best baseline

Full results → Appendix A2

Quality vs. Retrieval Cost

OLMo-2-13B · HotpotQA



Runtime breakdown → Appendix A3

Do Both Stages Matter?

OLMo-2-7B · 2WikiMultihopQA

Configuration	EM (%)		Retrievals
Full QuCo-RAG		32.7	2.61
Without initial check		30.2	1.82
Without runtime check		27.6	0.76

Removing runtime verification: -5.1 EM points

Beyond Short-Answer QA

OLMo-2-7B · Same statistical corpus, different tasks

ASQA · Long-form QA

28.5 DR

+2.0 over best baseline

1.72 retrievals / question

PubMedQA · Biomedical QA

66.4% accuracy

+3.0 points over best baseline

0.93 retrievals / question

Co-occurrence Is Not Entailment

A useful risk signal—not a truth test

Wrong relations

Same entities

Different factual relation

Lexical mismatch

Different names

Aliases can trigger
retrieval

Static knowledge

New entities

Old index, new facts

Zero count ≠ proven false · Positive count ≠ proven true

Failure analysis → Appendix A6

QuCo-RAG: Key Takeaways

Use corpus evidence to decide when more evidence is needed

Before generation

During generation

After retrieval

Rare entities?

Unseen pairs?

Regenerate.

RAGAS evaluates quality. QuCo-RAG controls retrieval.



Building RAG Agents with LLMs

Appendix

Jae Y. CHOI

NVIDIA DLI Certified Instructor & University Ambassador

Appendix A1: Main Results

OLMo-2 model scales

Table 1: Performance comparison on multi-hop QA benchmarks across OLMo-2 model scales. **Bold**: best; underline: second-best. *Improv.*: absolute gain over best baseline. 2Wiki: 2WikiMultihopQA.

Method	OLMo-2-7B				OLMo-2-13B				OLMo-2-32B			
	2Wiki		HotpotQA		2Wiki		HotpotQA		2Wiki		HotpotQA	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Wo-RAG	20.1	26.4	22.6	31.6	28.5	34.5	24.4	33.6	33.3	40.3	22.0	31.3
SR-RAG	23.7	30.7	<u>29.7</u>	<u>40.7</u>	28.9	35.7	<u>29.7</u>	<u>39.5</u>	<u>37.4</u>	<u>46.5</u>	29.5	40.4
FS-RAG	21.1	28.3	14.5	20.7	28.8	35.1	14.6	21.9	34.6	41.0	13.9	19.5
FLARE	22.9	28.9	20.3	28.4	26.2	31.5	15.3	21.9	32.0	39.3	28.3	39.8
DRAGIN	22.8	29.0	17.5	24.7	28.5	33.9	19.5	27.6	33.3	40.2	17.7	24.3
ETC	23.4	29.8	25.1	34.7	<u>29.7</u>	<u>35.9</u>	29.3	<u>39.5</u>	36.0	43.6	<u>30.8</u>	<u>42.2</u>
SeaKR	<u>25.3</u>	<u>32.7</u>	24.8	35.0	29.6	34.6	26.2	37.3	30.2	38.2	28.7	40.4
QuCo-RAG	32.7	41.1	35.3	46.1	41.7	49.1	35.0	46.8	46.8	56.2	41.6	54.2
<i>Improv.</i>	+7.4	+8.4	+5.6	+5.4	+12.0	+13.2	+5.3	+7.3	+9.4	+9.7	+10.8	+12.0

Appendix A2: Model Transfer

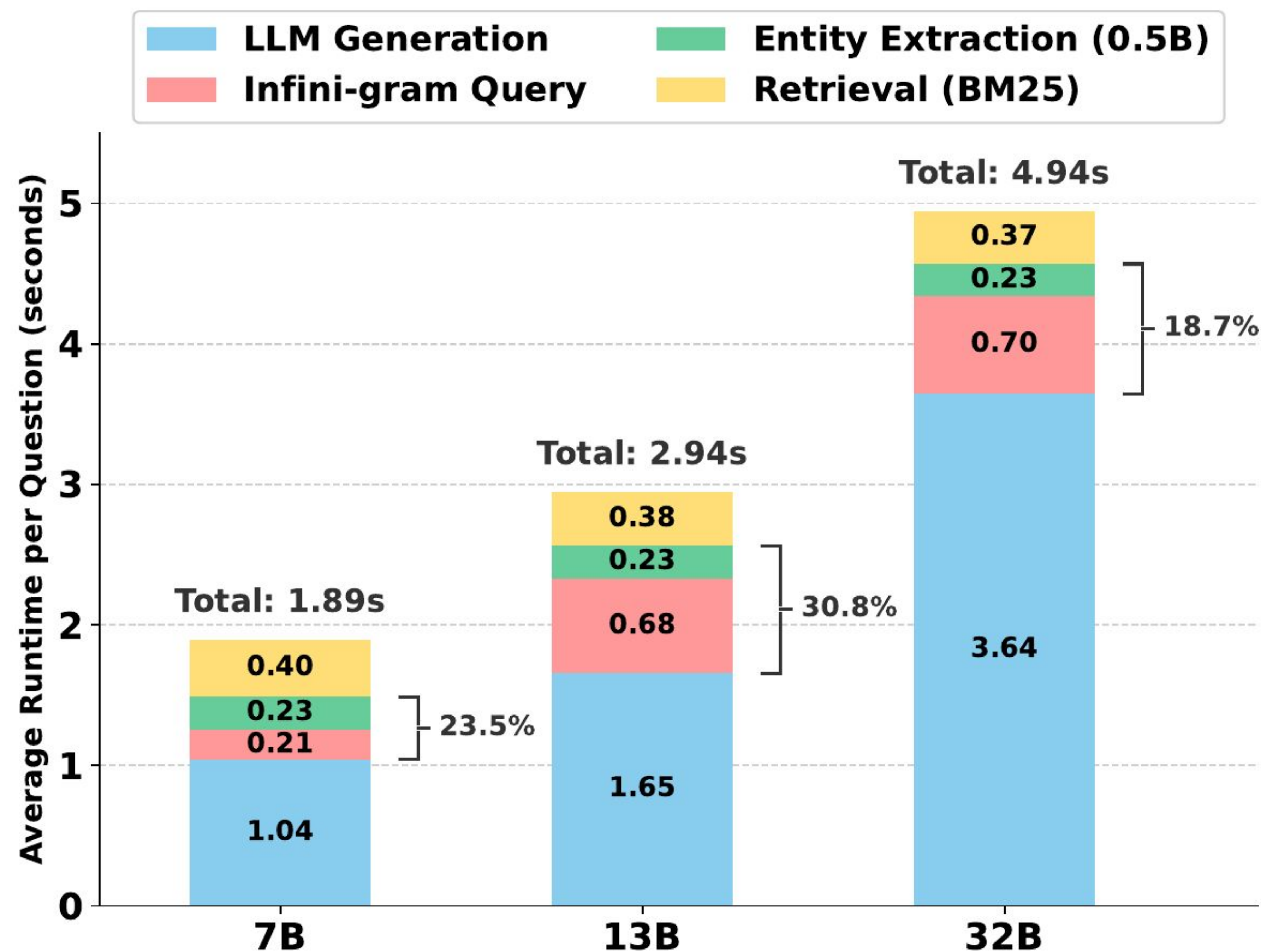
EM scores

Method	Qwen2.5-32B		Llama-3-8B	
	2Wiki	HPQA	2Wiki	HPQA
Wo-RAG	26.4	21.6	29.5	20.3
SR-RAG	23.0	31.0	12.9	22.7
FS-RAG	<u>35.9</u>	<u>38.6</u>	28.8	27.0
FLARE	26.4	24.1	26.6	22.2
DRAGIN	28.8	22.2	27.9	20.0
ETC	31.5	21.7	29.9	24.1
SeaKR	22.4	26.7	<u>33.5</u>	<u>33.5</u>
QuCo-RAG	50.0	41.6	38.4	36.2
<i>Improv.</i>	+14.1	+3.0	+4.9	+2.7

Method	GPT-4.1		GPT-5-chat	
	2Wiki	HPQA	2Wiki	HPQA
Wo-RAG	54.7	<u>40.1</u>	50.1	37.7
SR-RAG	<u>60.0</u>	38.8	<u>51.0</u>	<u>42.9</u>
FS-RAG	59.5	25.9	47.3	19.0
FLARE	49.8	38.7	-	-
Web-Tool	42.9	8.9	48.3	19.8
QuCo-RAG	64.6	48.2	59.7	48.4
<i>Improv.</i>	+4.6	+8.1	+8.7	+5.5

Appendix A3: Runtime

2WikiMultihopQA



Detection has a cost

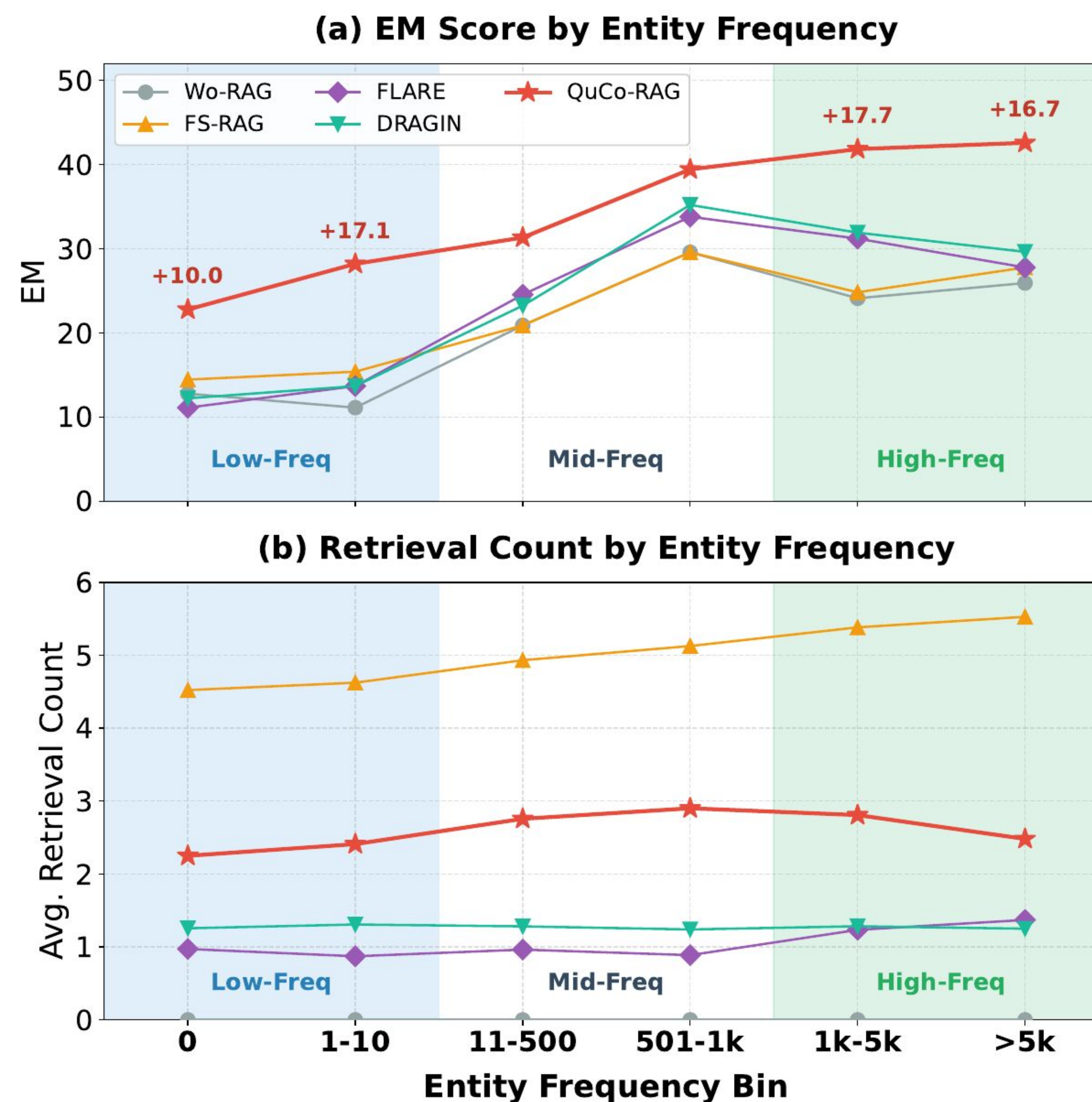
LLM generation still dominates runtime.

Local index: ~28 TB

Hosted API used in the study

Appendix A4: Entity Frequency

OLMo-2-7B, 2WikiMultihopQA



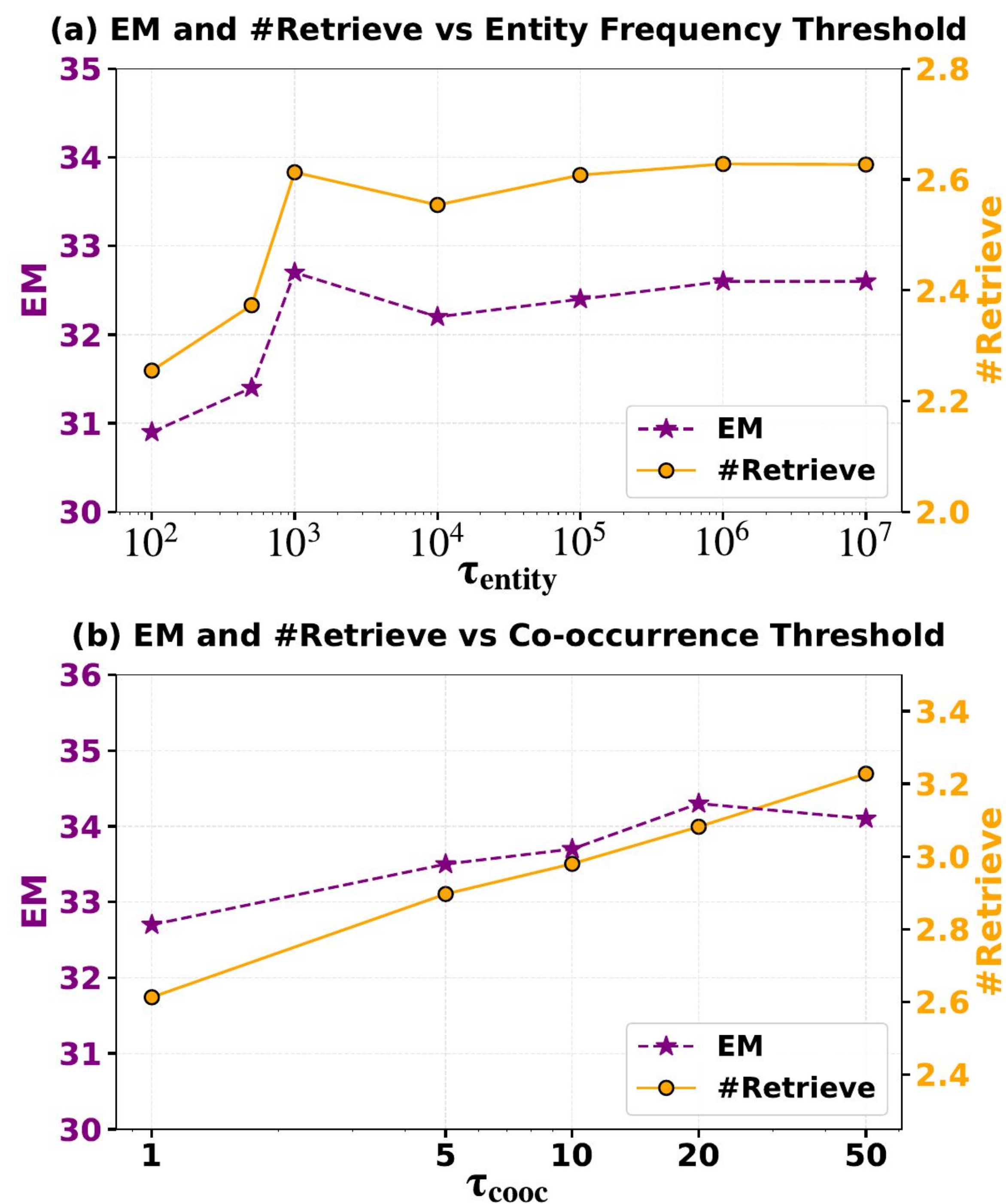
Long-tail gains

The largest gaps appear where entities are rare.

Frequency is a proxy for knowledge coverage.

Appendix A5: Thresholds

OLMo-2-7B, 2WikiMultihopQA



$$\tau_{\text{entity}} = 10^3$$

$$\tau_{\text{cooc}} = 1$$

Window: 1,000 tokens

Stricter checks → more retrieval

Appendix A6: Relation-Aware Check

Optional extension—not the default QuCo-RAG policy

	HotpotQA	2Wiki
Hallucinated sent.	183	202
Detected by S2 (TP)	95 (51.9%)	95 (47.0%)
Missed by S2 (FN)	88 (48.1%)	107 (53.0%)
Wrong relation in FN	60 (68.2%)	63 (58.9%)
Detection w/ + Rel. Check	149/183 (81.4%)	156/202 (77.2%)

Table 15: Post-hoc false negative analysis on 200 incorrect predictions per dataset (OLMo-2-7B). S2: Stage 2; TP: hallucination detected; FN: hallucination missed. The last row shows detection rate after adding the relation-aware extension.

More detection, more retrieval

200 incorrect predictions per dataset

Method	2WikiMultihopQA			HotpotQA		
	EM	F1	#Ret.	EM	F1	#Ret.
QuCo-RAG	31.5	39.8	2.43	35.4	46.5	2.70
+ Rel. Check	33.6	41.6	3.68	37.8	47.9	3.74
Δ	+2.1	+1.8	+1.25	+2.4	+1.4	+1.04

Table 16: End-to-end ablation of the relation-aware extension (OLMo-2-7B).



Building RAG Agents with LLMs

QuCo-RAG: Quantifying Uncertainty from the Pre-training Corpus
for Dynamic Retrieval-Augmented Generation

Jae Y. CHOI

NVIDIA DLI Certified Instructor & University Ambassador

본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

Copyright © Jae Y. CHOI (jaeyeong2022@gmail.com).

This material may be freely used, provided that the author and source are properly credited.